

Identification of Binding Sites on Human Serum Albumin for Somapacitan, a Long-Acting Growth Hormone Derivative

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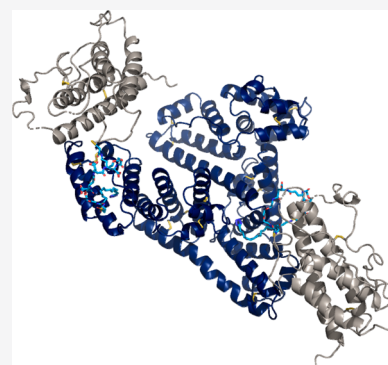
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ABSTRACT: Somapacitan, a human growth hormone derivative that binds reversibly to albumin, was investigated for human serum albumin (HSA) and HSA domain binding. Isothermal titration calorimetry (ITC) binding profiles showed high-affinity binding (~ 100 – 1000 nM) of one somapacitan molecule and low-affinity binding (~ 1000 – 10000 nM) of one to two somapacitan molecules to HSA. The high-affinity site was identified in HSA domain III using size exclusion chromatography (SEC) and ITC. SEC studies showed that the neonatal Fc receptor shields one binding site for somapacitan, indicating its position in domain III. A crystal structure of somapacitan in complex with HSA optimized for neonatal Fc receptor binding, having four amino acid residue replacements, identified a low-affinity site in fatty acid-binding site 6 (domain II). Surface plasmon resonance (SPR) showed these replacements affect the kinetics of the high-affinity binding site. Furthermore, small-angle X-ray scattering and SPR brace two somapacitan-binding sites on HSA.



Human serum albumin (HSA) has proven to be a versatile carrier for therapeutic proteins and peptides used to treat primarily endocrine and cancer diseases, by improving their pharmacokinetic properties. The long half-life of circulating albumin in human blood is a result of low renal clearance combined with pH-dependent endosomal rescue from degradation, mediated by the formation of a complex with the neonatal Fc receptor (FcRn)^{1,2} (Figure 1a).

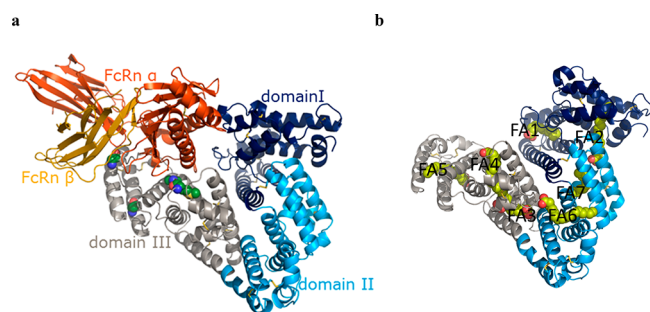


Figure 1. Illustration of HSA:FcRn interactions and fatty acid-binding sites in HSA. (a) FcRn-binding optimized HSA (HSAopt)–FcRn crystal structure from ref 5. HSAopt domains are colored as follows: domain I in dark blue, domain II in turquoise, domain III in gray, and mutated amino acid residues in green as space-filling spheres. FcRn is colored with a red heavy chain and a yellow $\beta 2$ -microglobulin chain. (b) HSA crystal structure (colored as in panel a) in complex with stearic acid in yellow space-filling spheres from ref 6. The fatty acids are numbered according to the prevalent terminology.

Noncovalent binding to albumin or fusion to albumin has been shown to extend the half-life of short-lived peptides and proteins.³ Fatty acid analogues and derivatives of human insulin and human glucagon-like peptide 1 such as insulin detemir (Levemir), insulin degludec (Tresiba), liraglutide (Victoza), and semaglutide (Ozempic) bind noncovalently to fatty acid-binding sites on HSA (Figure 1b) and thereby extend their half-lives.⁴

Somapacitan is a human growth hormone (GH) derivative that binds reversibly but extensively ($>99\%$) to endogenous circulating albumin.⁷ It is in development for once-weekly treatment of GH deficiency (GHD) in children and adults and for short children born small for gestational age (SGA). Somapacitan is a derivative with a single amino acid substitution in the backbone, to which a side chain containing a charged linker, tetrazole, and a fatty acid moiety with noncovalent albumin binding properties has been attached (Figure 2). Research into extending the half-life of GH has resulted in new long-acting GH derivatives that potentially seek to reduce treatment burden and improve patient convenience and the clinical benefit compared to daily GH treatment.^{8–10}

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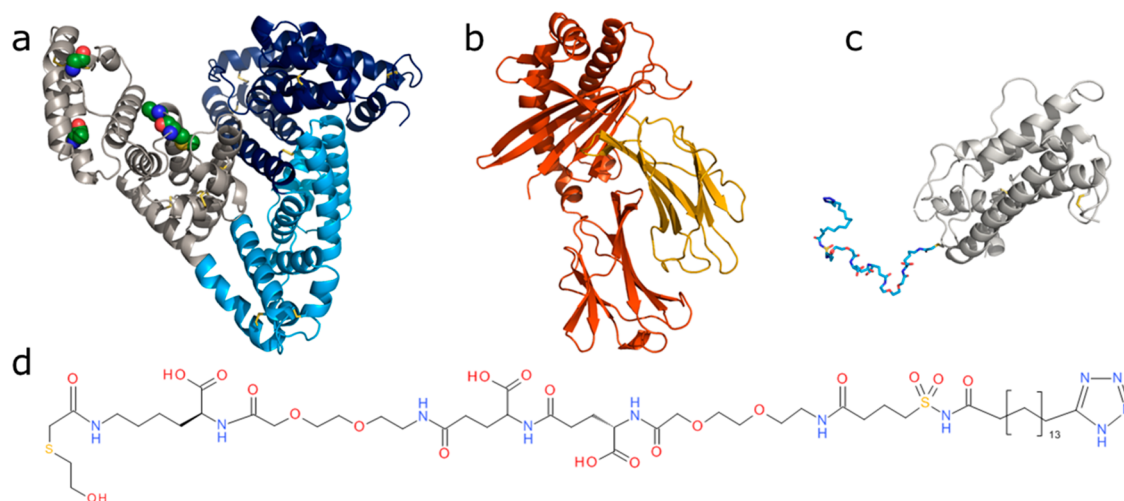


Figure 2. Key molecules used in the studies. Cartoons a–c are comparable in size. (a) For HSAopt, domains are colored as follows: domain I in dark blue, domain II in turquoise, domain III in gray, and mutated amino acid residues in green as space-filling spheres. (b) For FcRn, colored with a red heavy chain and a yellow β 2-microglobulin chain. (c) Somapacitan model with the growth hormone in light gray and the albumin-binding side chain shown as turquoise sticks. (d) Albumin-binding side chain molecular structure, where the thiol group of the engineered cysteine residue in the GH part of somapacitan, which is used for attachment, has been replaced by mercaptoethanol.

The aim of this study is to further clarify the mechanism of action for extension of the somapacitan plasma half-life by identifying specific binding sites on HSA for somapacitan and elucidating the binding stoichiometry and the binding affinity for those sites. This is not a straightforward task as HSA has at least seven fatty acid-binding sites, named FA1–FA7 (Figure 1b). A combination of several different methods has therefore been applied to illuminate intricacies of binding of somapacitan to HSA.

Crystal structures of HSA and a variant of HSA in complex with human FcRn have previously been described by Schmidt et al. and Oganessian et al.^{5,11} In the study presented here, the complex of the HSA variant and FcRn has been used as a tool to investigate the binding of somapacitan to HSA and to identify specific binding sites for somapacitan.

Furthermore, surface plasmon resonance (SPR) and isothermal titration calorimetry (ITC) techniques were used to study the interaction between somapacitan and HSA. SPR is a method used for monitoring molecular interactions in real time, without the requirement of labeled molecules for detection, and thereby characterizes both association and dissociation events of a particular interaction. SPR has previously been used to study the binding of proteins and peptides to HSA.¹² However, as a result of the multiple fatty acid-binding sites on HSA, only a qualitative analysis of binding of somapacitan to HSA was performed in the study presented here. ITC measures the change in heat that occurs in all associations. By using ITC, it is possible to study interactions in solution without immobilization. Previously, ITC has been used successfully to study the interaction of fatty acids and related molecules to serum albumins.^{13,14} Thus, it was possible to determine binding affinity, stoichiometry, and thermodynamic parameters for the binding of somapacitan to HSA.

MATERIALS AND METHODS

Somapacitan and Protected Albumin-Binding Side Chain. Somapacitan and the activated albumin-binding side chain of somapacitan were drug substance quality and obtained

from the production line at Novo Nordisk (Novo Nordisk, Denmark). The activated albumin-binding side chain (15.5 mg) was dissolved in 25 mM HEPES (pH 7.5). Ten milliliters of 0.5 M sodium chloride and 25 μ L of mercaptoethanol were added and left for 20 h at room temperature for protection of the activated albumin-binding side chain. The reaction was confirmed by HPLC and LC-MS analysis, diluted to 20 mL, and purified by preparative HPLC with a gradient of acetonitrile and 0.1% TFA from 35% to 75% against 0.1% TFA and water. The protected albumin-binding side chain was collected and lyophilized.

Cloning, Expression, and Purification of FcRn-Binding Optimized HSA (HSAopt) and FcRn. The DNA sequence encoding HSA with V418M, T420A, E505G, and V547A mutations (HSAopt), codon-optimized for expression in *Komagataella pastoris* (*Pichia pastoris*), was obtained by gene synthesis from GeneArt (Invitrogen, Thermo Fisher Scientific). A synthetic fragment containing the *Saccharomyces cerevisiae* mating factor α -1 prepro sequence fused to the HSA coding sequence was cloned into the pKO151 vector using restriction enzymes PacI and AgeI to generate pSUJ-154. Plasmid pSUJ-154 contains the constitutive glyceraldehyde-3-phosphate dehydrogenase promoter, PEP4 gene sequences for integration by homologous recombination into the PEP4 locus, and a synthetic Zeocin resistance selection gene. Approximately 2 μ g of pSUJ-154 was linearized with the restriction enzyme SapI prior to DNA transformation into *K. pastoris* strain CBS704 (obtained from Westerdijk Fungal Biodiversity Institute) using electroporation.¹⁵ Yeast transformants were selected on yeast extract peptone dextrose (YPD) and 100 μ g/mL Zeocin plates [1% yeast extract (BD Difco), 2% peptone (BD Difco), 2% agar (BD Difco), 2% glucose (BD Difco), and 100 μ g/mL Zeocin (Invitrogen)] and subsequently grown on YPD medium plates containing 3000 μ g/mL Zeocin to isolate strains with multicopy gene integration. Individual transformants were grown in liquid culture, and the strain secreting the largest amount of HSAopt protein, ySUJ-291, was used for further studies.

A single colony from the YPD medium plate of ySUJ-291 was selected and used for inoculating a shake flask containing a

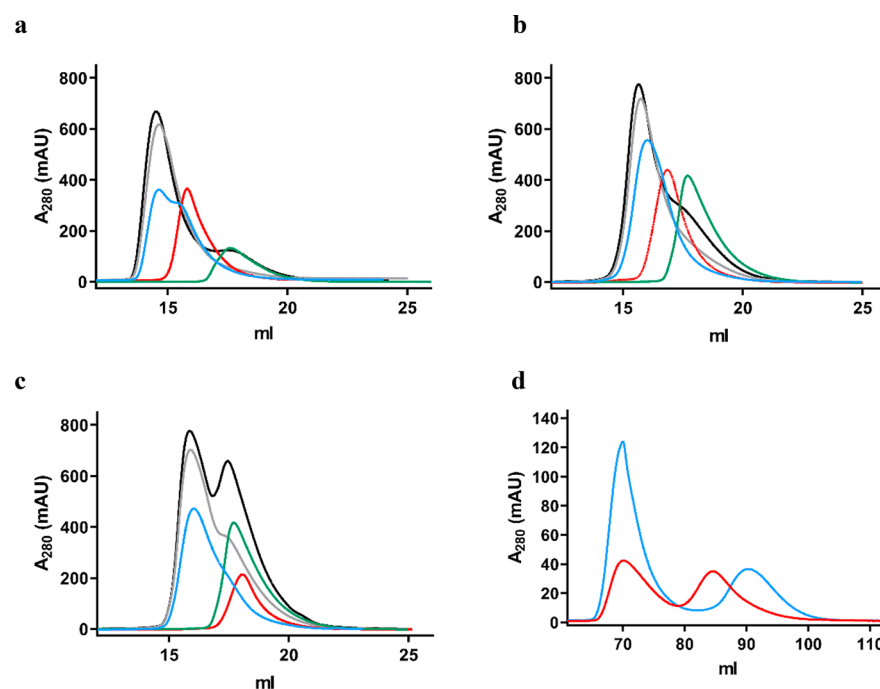


Figure 3. SEC on HSA and HSA domains in complex with somapacitan and FcRn. (a) Superposed chromatograms of the HSA:somapacitan complex in a 1:1 molar ratio (light blue), a 1:2 molar ratio (gray), and a 1:3 molar ratio (black) with somapacitan alone (green) and HSA alone (red). (b) Superposed chromatograms of HSA domain I+II: somapacitan in 1:1 (light blue), 1:2 (gray), and 1:3 (black) molar ratios with somapacitan alone (green) and domain I+II alone (red). (c) Superposed chromatograms of the HSA domain III:somapacitan complex in 1:1 (light blue), 1:2 (gray), and 1:3 (black) molar ratios with somapacitan alone (green) and domain III alone (red). (d) Superposed chromatograms of the HSA:somapacitan complex in a 1:4 molar ratio (blue) and the HSA:somapacitan:FcRn mixture in a 1:2:1 molar ratio (red).

defined minimal medium supplemented with yeast extract containing glucose as the source of carbon. The inoculum was incubated for 2 days in an orbital shaker heated to 28 °C and then used to inoculate a 2.5 L Braun laboratory fermentor containing a yeast minimal medium supplemented with yeast extract. The yeast was propagated as a three-step culture consisting of an initial batch phase followed by a fed-batch phase and finally an aerobic continuous culture. For the fed batch, the culture was fed with a glucose solution and during the continuous cultivation phase with the growth medium supplied with glucose giving a constant growth rate of 0.08 h⁻¹. The total volume was 1.5 L, and the culture was maintained at 26 °C and pH 6.2 during the entirety of the experiment. The culture broth was collected cold (5–10 °C) during the 180 h continuous phase. The collected culture broth was harvested by batch centrifugation (6000g for 30 min).

Supernatant from continuous culture expression of HSAopt from *K. pastoris* was collected, and the pH adjusted to 5.5 by addition of sodium hydroxide. Five liters of medium was loaded on a 90 mL AlbuPure column (Novozymes) previously equilibrated with 50 mM sodium citrate, pH 5.5 buffer. The column was washed with buffer A [50 mM ammonium acetate (pH 7.0)] and eluted with a gradient of buffer B [50 mM ammonium acetate and 100 mM sodium octanoate (pH 7.0)]. HSAopt-containing fractions were concentrated and made fatty acid free as described by Chen et al.¹⁶ The defatted HSAopt was subjected to size exclusion chromatography (SEC) on a Superdex 200 26/60 column equilibrated with a 50 mM sodium phosphate, 100 mM sodium chloride, pH 7.4 buffer.

Human FcRn ectodomain was composed of two components, the α -chain and β -chain (hB2M). Plasmids encoding the

α -chain and β -chain were maxi prepped with the QIAGEN Endotoxin free Max prep kit (QIAGEN Sciences) and transfected into HEK293-6E cells. The FcRn culture supernatant was harvested after being cultured for 6 days in Freestyle 293 medium with Fluoronic F68 and Geneticin (Invitrogen) and centrifuged at 5000 rpm for 10 min before filtration. The filtered supernatant was applied to a nickel Sepharose excel column (GE Healthcare), equilibrated in 20 mM monosodium phosphate, 0.5 M sodium chloride, pH 7.4 buffer. The bound protein was eluted with the equilibration buffer containing 20 mM imidazole. The pooled fractions were further purified by SEC on a Superdex 75 pg column (GE Healthcare) in PBS. Purified human FcRn was sterilized by filtration through a 0.2 μ m filter unit (Sartorius). The purity was analyzed by sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) and size exclusion HPLC. The protein identity was confirmed by mass spectrometry.

FcRn was mixed with HSAopt in a 10% molar excess. The mixture was dialyzed against 100 mM MES, 100 mM sodium chloride, pH 6.5 buffer and run on a Superdex 200 26/60 column (GE Healthcare) equilibrated with the same buffer to isolate the 1:1 complex. This complex was further mixed with somapacitan in an 8-fold molar excess, and after incubation overnight, the complex was the subject for a new SEC experiment. The large peak of excess somapacitan was discarded, and the ternary FcRn:HSAopt:somapacitan complex was isolated. The same procedure was repeated for the isolation of the FcRn:HSAopt:albumin-binding side chain ternary complex.

HSA:Somapacitan Complex Formation Studied by SEC. Recombinant HSA (Novozymes, Denmark), made fatty acid free using the same treatment that was used for HSAopt,

was mixed with somapacitan in an equimolar ratios and in 2- and 3-fold molar excesses. After incubation overnight, the samples were subjected to SEC on a Superdex 200 10/300 GL column (GE Healthcare) equilibrated with a 100 mM MES, 140 mM sodium chloride, pH 6.5 buffer (Figure 3a).

SEC on HSA Domains and Somapacitan Complexes. Recombinant HSA domain III and domain I+II were purchased from Albumin Biosciences. Domain III alone and domain I+II were first run on a Superdex 200 10/300 GL column (GE Healthcare) equilibrated in 100 mM MES (pH 6.5) and 140 mM sodium chloride to remove the remaining dimers and impurities. Thereafter, the respective domain preparation was mixed with somapacitan in an equimolar ratio and in 2- and 3-fold molar excesses of somapacitan and incubated overnight, and SEC was performed again using the same column and buffer (Figure 3b,c).

SEC of the HSA:Somapacitan Complex Mixed with FcRn. Recombinant HSA (Novozymes, Denmark), made fatty acid free using the same treatment that was used for HSAopt, was mixed with somapacitan in a 4-fold molar excess. After incubation for 2 h, the samples were subjected to SEC on a HiLoad 16/60 Superdex 200 pg column (GE Healthcare) equilibrated with a 100 mM MES, 140 mM sodium chloride, pH 6.0 buffer (Figure 3d). The HSA:somapacitan complex peak was collected and mixed in a 1:1 molar ratio with FcRn, incubated for 2 h, and rerun on the same SEC column and buffer (Figure 3d). The fractions were analyzed by SDS-PAGE for identification of the protein peaks in the chromatograms.

Crystallization and Diffraction Data Collection. Crystallization was performed using the vapor diffusion sitting drop technique mixing drops of 360 nL of protein and 360 nL of precipitant and equilibration over 80 μ L of precipitant at 20 °C. The precipitant used for the 18.4 mg/mL FcRn:HSAopt:somapacitan ternary complex consisted of 0.2 M ammonium acetate, 0.1 M sodium acetate (pH 5), and 20% (w/v) PEG4000 and for the 14.5 mg/mL FcRn:HSAopt:albumin-binding side chain complex, 0.1 M sodium cacodylate (pH 5.5) and 20% (w/v) PEG3350. For both crystals, the protein used was in the same buffer as the running buffer for SEC (100 mM MES, 100 mM sodium chloride, pH 6.5). The crystals were cryo-protected in precipitant containing 20% ethylene glycol and thereafter cryo-cooled in liquid nitrogen before being shipped to Swiss Light Source under cryogenic conditions. Diffraction data were collected at beamline X06DA on a Pilatus 2M detector (Dectris) at 100 K. The data were autoindexed, integrated, and scaled with programs in the XDS suite.¹⁷ Both ternary complexes belong to the same crystal form, and the statistics are summarized in Table 1.

Structure Determination and Refinement. The structures were determined by molecular replacement with Phaser¹⁸ as implemented in Phenix.¹⁹ The structure of PDB entry 4K71⁵ was used as the search model for the FcRn:HSAopt:somapacitan ternary complex, and the refined model of this ternary complex stripped from ligands was in turn used as the search model for the albumin-binding side chain ternary complex. The parts of somapacitan and the albumin-binding side chain that were visible in the difference electron density maps were model built using COOT,²⁰ and rounds of refinement in Phenix and model building in COOT were applied. Difference electron density was also found for a MES buffer molecule at the border between fatty acid-binding sites 3 and 4 and a cysteine residue covalently bound to the free

Table 1. Diffraction Data and Refinement Statistics^a

	FcRn:HSAopt:somapacitan	FcRn:HSAopt:albumin-binding side chain
wavelength (Å)	1.0000	1.0000
resolution range (Å)	48.85–1.95 (2.02–1.95)	44.14–2.45 (2.54–2.45)
space group	<i>P</i> 2 ₁ 2 ₁ 2 ₁	<i>P</i> 2 ₁ 2 ₁ 2 ₁
unit cell dimensions	75.29 Å, 108.12 Å, 159.53 Å, 90°, 90°, 90°	77.02 Å, 114.45 Å, 159.35 Å, 90°, 90°, 90°
total no. reflections	637260 (59717)	347785 (33301)
no. of unique reflections	95478 (9430)	52520 (5168)
multiplicity	6.7 (6.3)	6.6 (6.4)
completeness (%)	99.91 (99.98)	99.91 (99.90)
$\langle I \rangle / \sigma(I)$	11.23 (1.25)	17.08 (1.12)
Wilson <i>B</i> -factor (Å ²)	27.05	58.05
<i>R</i> _{merge}	0.1506 (1.494)	0.08365 (1.683)
<i>R</i> _{meas}	0.1634 (1.628)	0.09086 (1.831)
<i>R</i> _{pim}	0.06289 (0.6438)	0.03514 (0.7175)
CC _{1/2}	0.997 (0.464)	0.999 (0.585)
CC*	0.999 (0.796)	1 (0.859)
no. of reflections used in refinement	95436 (9430)	52497 (5165)
no. of reflections used for <i>R</i> _{free}	4772 (464)	2623 (274)
<i>R</i> _{work}	0.1823 (0.3091)	0.2160 (0.4185)
<i>R</i> _{free}	0.2262 (0.3485)	0.2583 (0.4339)
CC(work)	0.959 (0.719)	0.944 (0.744)
CC(free)	0.947 (0.682)	0.931 (0.689)
no. of non-hydrogen atoms	8657	7722
macromolecules	7685	7598
ligands	94	98
solvent	878	26
protein residues	951	951
root-mean-square deviation (bonds)	0.012	0.011
root-mean-square deviation (angles)	1.44	1.69
Ramachandran favored (%)	97.35	96.71
Ramachandran allowed (%)	2.12	2.44
Ramachandran outliers (%)	0.53	0.85
rotamer outliers (%)	0.24	0.12
Clashscore	5.98	6.52
average <i>B</i> -factor (Å ²)	35.10	71.08
macromolecules	34.40	71.43
ligands	40.43	47.73
solvent	40.60	58.32

^aStatistics for the highest-resolution shell are shown in parentheses.

cysteine residue Cys48 of the FcRn α -chain, which both were added and refined. Refinement statistics are listed in Table 1. The final models and structure factors are deposited in the PDB as entries 6QIO and 6QIP for the somapacitan and somapacitan albumin-binding side chain ternary complexes, respectively.

Small-Angle X-ray Scattering (SAXS) Data Collection and Analysis. SAXS data were collected on the following SEC-purified complexes: FcRn:HSAopt:somapacitan (1:1:1), FcRn:HSAopt:albumin-binding side chain (1:1:1), and re-

Table 2. SAXS Data

	FcRn:HSAopt:somapacitan	FcRn:HSAopt:albumin-binding side chain	HSA:somapacitan
molar ratio	1:1:1	1:1:1	1:2
protein concentration (mg/mL)	4.5	5	6.3
X-ray source	MAX-lab I911-4	MAX-lab I911-4	MicroMax-007 HF (Rigaku)
wavelength (Å)	1.0000	1.0000	1.54187
exposure time (s)	120	120	3600
temperature (°C)	20	20	20
R_g (Å)	42	36	41
D_{max} (Å)	147	126	139
Porod volume (Å ³)	227351	191760	191616
molecular mass from Porod volume (Da)	132	113	113
theoretical molecular mass (Da)	133	111	113
SASDB accession code	SASDF22	SASDF32	SASDF42

combinant HSA:somapacitan (1:2). The first two scattering curves were measured at MAX-lab, Lund university, Sweden, beamline I911-4 on a Pilatus 1M detector from Dectris, and the third data set was collected on a BioSAXS-2000 instrument mounted on a MicroMax-007 HF rotating anode (Rigaku) equipped with a Pilatus 100K detector from Dectris. Experimental conditions are described in Table 2. The SAXS curves were buffer subtracted and analyzed with the SAS Data analysis module from the ATSAS 2.5.0 package,²¹ and the deduced values are found in Table 2. A rigid body refinement using a HSA model in complex with octadecanoic acid⁶ and two growth hormone molecules²² was applied in the ATSAS SASREF module²³ using 15 Å distance constraints from residue 101 of growth hormone to residues 209 (FA6) and 502 (FAS) of HSA. The SAXS data and models are deposited in the small-angle scattering biological data bank (<https://www.sasbdb.org/>) with the accession codes listed in Table 2.

SPR Analysis. SPR analysis was used to evaluate the kinetics of binding of somapacitan to albumin. The analyses were performed on a Biacore T200 instrument from GE Healthcare with a C5 sensor chip. The running buffer consisted of 50 mM MES, 150 mM sodium chloride, 3 mM EDTA, and 0.05% Tween P20 (BR-10000-54) at pH 6.3. Immobilization of anti-hGH was performed at 25 °C, and the binding analysis was performed at 37 °C. An amine coupling kit from GE Healthcare (BR-1006-33) was used for amine coupling of the capturing antibody, anti-hGH Fitzgerald 10-G05B (clone 94169, batch 7334). The capturing antibody was immobilized in both the reference and active flow cells at 25 µg/mL, in 10 mM sodium acetate (pH 5.0), resulting in approximately 1000 RU of immobilized antibody in several experiments. Aggregates were removed from HSA (A9511 Sigma) by SEC, and a stock solution of 1.56 mg/mL in PBS (pH 7.4) was used for further dilution in running buffer. HSAopt was diluted in running buffer. For binding analysis, somapacitan was captured in the active flow cell at a concentration of 3 nM. This typically resulted in final capture levels of hGH of 70–100 RU. Binding of HSA and HSAopt was conducted by injecting five different concentrations of HSA (0, 64, 320, 640, and 3200 nM) in triplicate over both flow cells to allow for comparative analyses of binding relative to binding to the reference flow cell. The sensor chip surface was regenerated after each injection cycle with 3 M magnesium chloride. This regeneration step removed the somapacitan and any bound HSA from the immobilized capture antibody surface and allowed for the subsequent binding of the next interaction sample pair. The regeneration procedure did not

remove the directly immobilized anti-GH capture antibody from the chip surface. All instrument control, analysis, and calculations were performed using the software provided by GE Healthcare (Biacore T200). Binding curves were processed by double referencing (subtraction of reference surface signals as well as blank buffer injections over captured somapacitan). This allowed correction for instrument noise, bulk shift, and drift during sample injections. As a result of the multiple fatty acid-binding sites on HSA, it was not possible to obtain reliable kinetic parameters; hence, the analysis was only qualitative.

ITC. Titration experiments were performed using a PEAQ-ITC instrument (Malvern). In the titration experiments performed in this study, 0.7 µL aliquots of a 460 µM somapacitan solution were titrated into a 13 µM HSA solution and the heat flow was monitored. The stepwise addition of somapacitan resulted in gradual saturation of HSA, and ultimately, all injected somapacitan remained unbound. Thus, at saturation only small changes in heat flow were observed as a result of mixing of somapacitan and HSA solutions. Titrations were carried out at selected temperatures between 13 and 37 °C. The data were evaluated using the MicroCal PEAQ-ITC Evaluation software (Malvern), and integrated heat signals were fitted to “one set of sites” or “two sets of sites” models to obtain the binding enthalpy, affinity, and stoichiometry estimates. Overall, the data evaluation was carried out as previously described by Nielsen et al.¹³

RESULTS

Preparation and Characterization of Somapacitan: HSA Complexes. The FcRn-binding optimized HSA (HSAopt) used for the crystal structures in this study has the same amino acid modifications as that used by Schmidt et al.,⁵ i.e., V418M, T420A, E505G, and V547A. It was recombinantly produced in *K. pastoris* (*P. pastoris*) and purified by affinity chromatography, and bound fatty acid was removed by active charcoal treatment at low pH.¹⁶ FcRn was produced by recombinant expression in human embryonic kidney cells. FcRn is a non-covalently bound heterodimer consisting of an MHC class I-like heavy chain and a β2-microglobulin light chain, which is the light chain for all MHC class I molecules. In the variant used here, the β2-microglobulin light chain has a C-terminal histidine tag for affinity purification. Structures of somapacitan, HSAopt, FcRn, and the albumin-binding side chain of somapacitan used in the crystallographic and SAXS studies are shown in Figure 2. FcRn:HSAopt:somapacitan and FcRn:HSAopt:albumin-binding side chain ternary complexes were isolated by size

exclusion chromatography (SEC). The sizes of both complexes were verified by SAXS analysis and found to correspond well to those of the ternary complexes (Table 2).

Structure. Both ternary complexes (FcRn:HSAopt:somacipatan and FcRn:HSAopt:albumin-binding side chain) crystallized in the same crystal form, and the crystal structures show clear difference electron density at fatty acid-binding site 6 (FA6) between HSA subdomains IIA and IIB (Figure 4). For

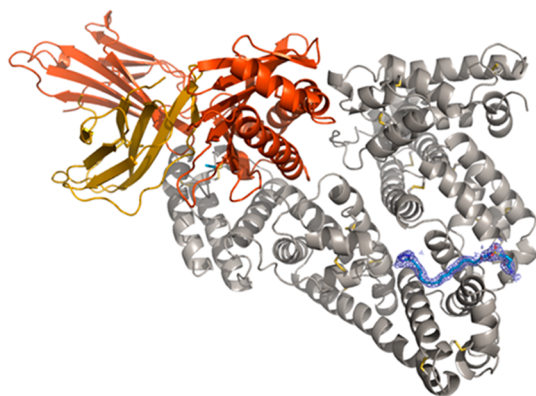


Figure 4. HSAopt:FcRn:somacipatan ternary complex. Cartoons of FcRn (red and yellow) and HSAopt (gray) and a stick representation of the ordered part of the somacipatan albumin-binding side chain are shown in turquoise with its $2F_o - F_c$ difference electron density map at the 1σ level in steel blue.

both the somacipatan and the albumin-binding side chain complexes, there is no difference electron density beyond the sulfonimide C4 part of the linker and hence the two different complexes result in analogous crystal structures. The presence of somacipatan in the crystals was verified by dissolving the crystal after diffraction data collection. A silver-stained SDS-PAGE gel unambiguously confirmed the presence of somacipatan (data not shown). In the crystal, the molecules are packed such that there is a large space for the GH and the linker part of the albumin-binding side chain.

The nature of binding of the fatty acid to HSA, including FA6, has previously been studied by X-ray crystallography, where C10, C12, C14, C16, and C18 fatty acids have been co-crystallized with recombinant HSA and their structures determined⁶ (Figure 5). That study showed that at FA6, there are no amino acid residues making specific contacts with the carboxylate headgroup of any of these fatty acids and that their binding positions vary depending on the fatty acid chain length. For example, the C10 length fatty acid binds at two positions adjacent to each other in this pocket, while the C12–C18 fatty acids bind at only one position in FA6. The placement of the tetrazole headgroup of the albumin-binding side chain is similar to the C14 to C18 fatty acids in the study by Bhattacharya et al.⁶

As part of our study, we tried to form the 1:1 complex between unmodified, recombinant HSA and somacipatan by mixing the components in a 1:2 HSA:somacipatan molar ratio with the intention to remove the somacipatan excess by SEC. However, no excess somacipatan was present in the 1:2 mixture, but in a 1:3 molar ratio, an excess of somacipatan was found (Figure 3a). This suggests that FcRn is blocking a binding site for somacipatan on HSA. The same experiments were repeated with HSAopt and showed the same feature (data not shown). There is no unambiguous way to conclude

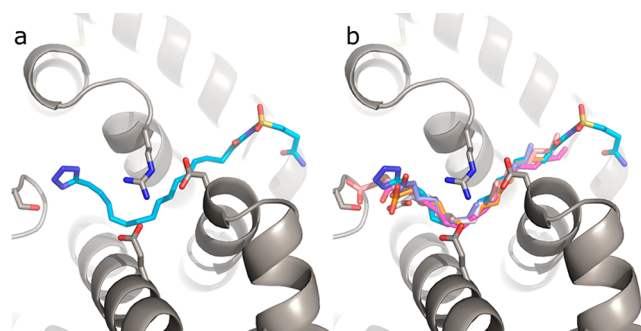


Figure 5. Close-up of HSA FA6. The cartoon of HSAopt is colored gray with the side chains of amino acid residues Arg209, Asp324, Glu354, and Ser480 shown as sticks. (a) Visible part of the albumin-binding side chain from the somacipatan ternary structure displayed as turquoise sticks. (b) Superpositions with HSA crystal structures from ref 6 in complex with fatty acids of length C10 (pale pink), C12 (white), C14 (purple), C16 (orange), and C18 (magenta) displayed as sticks.

from the available crystal structures if the binding site blocked by FcRn is placed on domain I or III of HSA, because both of these domains interact with FcRn (Figure 1). The interacting surface of domain III is the largest, but FcRn binding also restrains movement of domain I. Therefore, we decided to investigate the binding of somacipatan to commercially available domain preparations of HSA. One preparation of HSA comprised residues 1–385, which corresponds to domains I and II (domain I+II), and another preparation of HSA comprised residues 381–585, which corresponds to domain III. The stoichiometries of HSA domain:somacipatan complexes were examined by SEC in a manner similar to that of the HSA:somacipatan complexes described above (Figure 3b,c). In addition to binding at domain I+II, likely to correspond to the site localized by the crystal resolved structure, SEC studies also showed binding of somacipatan to domain III. On the basis of this knowledge, it is possible to construct a model of HSA with somacipatan bound at two sites: one at FA6 in domain II, as determined from the crystal structures, and one at a fatty acid-binding site in domain III, where both FA4 and FA5 are possible binding sites, as deduced from SEC studies of full-length HSA and HSA domains. The positioning of GH relative to HSA was guided using rigid body refinement against the SAXS data of the 1:2 HSA:somacipatan complex, which resulted in a χ^2 value of 1.45. A cartoon of this model, including the albumin-binding side chain of the two somacipatan molecules bound in FA6 and FA5, is shown in Figure 6. The binding of somacipatan to FA5 was chosen as an illustration in Figure 6, but it is not possible from the data to distinguish if it is FA4 or FA5 that is the correct binding site for somacipatan.

A SEC experiment in which a preformed 1:2 HSA:somacipatan complex was mixed so that the HSA:somacipatan:FcRn molar ratio was 1:2:1 resulted in two separate peaks: one with HSA and somacipatan and one with FcRn alone (Figure 3d). This indicates that FcRn cannot compete to remove the somacipatan off HSA once bound.

Binding of Somacipatan to HSA. The interaction between somacipatan, captured on the sensor chip matrix by anti-GH, and HSA or HSAopt at five different concentrations was monitored by SPR at 37 °C. The overlay plot of individual sensorgrams (Figure 7) shows that HSA and HSAopt both bind to somacipatan in a concentration-dependent manner.

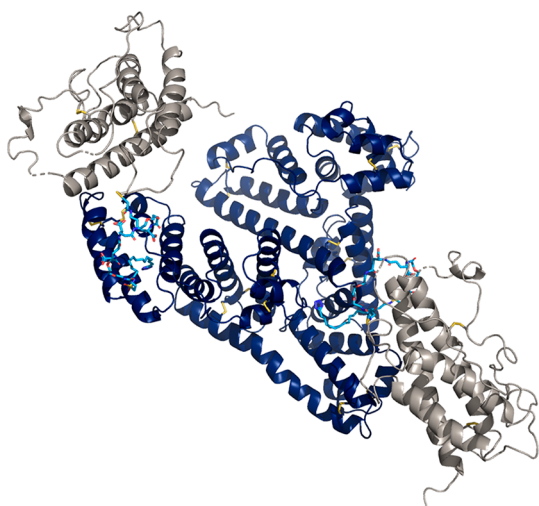


Figure 6. Two binding sites for somapacitan (gray) on HSA (dark blue). The albumin-binding side chains of FAS (on the left) and FA6 (on the right) are shown as turquoise sticks. The model has been created by rigid body fits of two GH molecules and one HSA molecule to SAXS data.

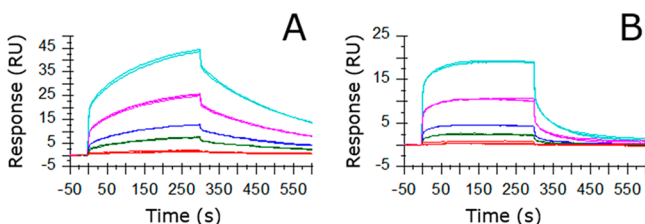


Figure 7. SPR analysis. Sensorgrams for a concentration series (3200, 1600, 640, 320, and 64 nM) of both (A) HSA and (B) HSAopt binding at 37 °C to 70–75 RU of somapacitan captured on a CM4 sensor via the covalently immobilized anti-GH antibody. This figure shows overlays of three replicates.

This interaction is more complex than a 1:1 binding relationship for the HSA variants tested and does not reach equilibrium for wild-type (wt) HSA, an observation that can be expected as a result of the presence of multiple fatty acid-binding sites. It was consequently not possible to obtain reliable kinetic parameters for the binding interaction when fitting it to the 1:1 Langmuir binding model. Although the binding kinetics of neither wt HSA nor HSAopt can be resolved, qualitative differences are clearly observed. The binding profile obtained by the combination of slow and fast kinetics is most dominant for HSA. The kinetics for HSAopt show a smaller contribution of slow kinetics, and equilibrium is reached for all sample concentrations. The kinetic profiles are similar for recombinant defatted HSA and serum-derived HSA (data not shown).

Titration profiles of the somapacitan:HSA complex at 13 to 37 °C using ITC are shown in Figure 8. Visual evaluation of the titration profiles confirms that somapacitan binds to HSA. Initially, one molecule of somapacitan binds to HSA with high affinity at all tested temperatures. At temperatures between 13 and 25 °C, this is followed by low-affinity binding of somapacitan molecules. Complete saturation is obtained at a molar ratio of 4 ([somapacitan]/[HSA] $\sim$ 4). A model of two classes of independent binding sites accounts well for the titration profiles obtained at 13–25 °C, and the binding parameters using the model are summarized in Table 3. At 31

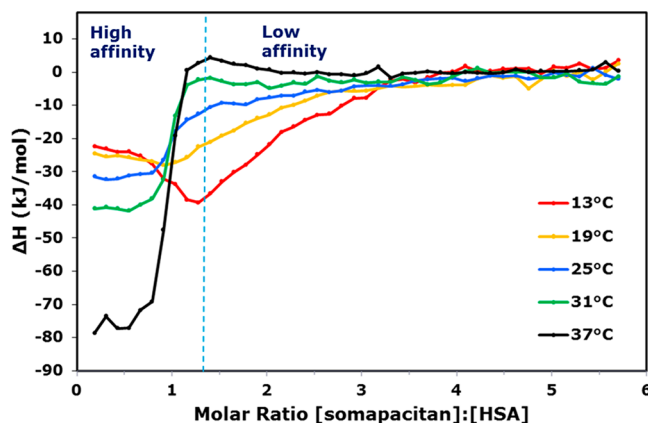


Figure 8. ITC titration experiments with HSA and somapacitan. Titration of 15 μ M HSA with 0.49 mM somapacitan in pH 7.1 buffer at 13, 19, 25, 31, and 37 °C.

and 37 °C, a “one-site binding model” is sufficient to account for the titration profiles. The high-affinity binding is in the range 10^{-7} to 10^{-8} M, whereas the low-affinity binding is on the order of 10^{-6} M. Evaluation of the binding parameters for the high-affinity site (≤ 31 °C) suggests that binding is mainly enthalpically driven, with an entropically favorable contribution. At 37 °C, the entropic contribution is unfavorable but this is compensated by a large favorable enthalpic contribution. For low-affinity binding, the change in binding free energy is comparable across the tested temperatures; however, an enthalpy–entropy compensation effect is observed. Comparison of the binding enthalpies reveals opposite temperature dependency for the high- and low-affinity sites corresponding to a negative change in heat capacity (ΔC_p) for the high-affinity site and a positive ΔC_p for the low-affinity binding sites. The results are indicative of profound differences in the binding mechanisms for the low- and high-affinity sites. The large negative change in ΔC_p suggests a binding dominated by hydrophobic interactions,¹³ whereas the positive ΔC_p for low-affinity binding indicates that forces other than just hydrophobic ones are involved in binding.

Binding of Somapacitan to Albumin Domains. To elucidate the location of the somapacitan-binding sites on albumin, ITC titrations were performed using protein constructs of HSA domain I+II and domain III. The titration profiles of somapacitan:domain I+II and somapacitan:domain III complexes at 37 °C are presented in Figure 9. The profile obtained for domain III is comparable to the profile of full-length HSA (Figure 8, 37 °C profile), which added to the evidence that the high-affinity binding site is located in domain III. A “one set of sites” binding model accounts well for the titration profile obtained, and the estimated binding affinity and enthalpy change are 1.9×10^{-7} M and -87.3 kJ/mol, respectively. The binding parameters for the domain III construct are comparable to the binding parameters obtained for full-length HSA at 37 °C (cf. Table 3). The profile of domain I+II indicates an interaction with somapacitan. SPR binding studies of HSA domains were not included because of the high level of unspecific binding to the reference flow cell.

DISCUSSION

In this paper, we have investigated the properties of binding of somapacitan to HSA. Somapacitan is a long-acting GH derivative, which binds in a noncovalent, reversible fashion

Table 3. Compilation of Thermodynamic and Stoichiometric Data for the Binding of Somapacitan to HSA

	13 °C	19 °C	25 °C	31 °C	37 °C
High-Affinity Sites					
ΔH (kJ/mol)	-26.0 ± 0.9	-23.2 ± 0.3	-32.4 ± 0.5	-39.6 ± 1.3	-80.7 ± 0.8
K_D (M)	$(4.5 \pm 1.3) \times 10^{-8}$	$(2.8 \pm 0.1) \times 10^{-8}$	$(1.9 \pm 0.3) \times 10^{-8}$	$(1.5 \pm 0.2) \times 10^{-8}$	$(1.3 \pm 0.1) \times 10^{-7}$
ΔG (kJ/mol)	-40.3 ± 0.8	-42.2 ± 0.1	-44.1 ± 0.4	-45.6 ± 0.4	-40.9 ± 0.2
$T\Delta S$ (kJ/mol)	13.3 ± 1.7	19.0 ± 0.4	11.7 ± 0.9	5.7 ± 1.7	-39.8 ± 1.0
stoichiometry (N)	1.0 ± 0.1				
Low-Affinity Sites					
ΔH (kJ/mol)	-48.2 ± 1.0	-28.6 ± 0.4	-12.1 ± 0.5	—	—
K_D (M)	$(4.0 \pm 0.1) \times 10^{-6}$	$(4.5 \pm 0.2) \times 10^{-6}$	$(8.1 \pm 0.6) \times 10^{-6}$	—	—
ΔG (kJ/mol)	-29.6 ± 0.1	-29.9 ± 0.1	-29.1 ± 0.2	—	—
$T\Delta S$ (kJ/mol)	-23.2 ± 1.1	1.4 ± 0.5	17.0 ± 0.7	—	—
stoichiometry (N)	2.2 ± 0.1	—	—	—	—

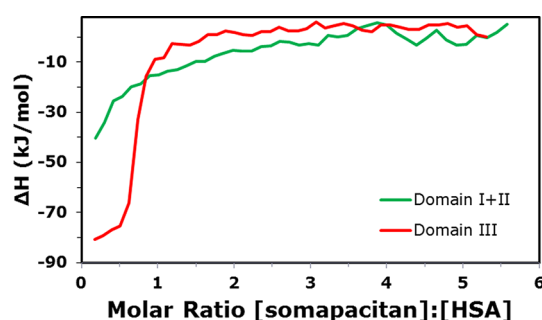


Figure 9. ITC titration experiments with HSA domains and somapacitan. Titration of 15 μ M HSA domain I+II and domain III with 0.49 mM somapacitan in pH 7.1 buffer at 37 °C.

to HSA via its C16 fatty acid-tetrazole-binding moiety. The protein binding has been assessed in plasma from several species, including humans, monkeys, minipigs, dogs, rats, rabbits, and mice, and showed a high level of binding of >99% across species. In clinical trials in children and adults with growth hormone deficiency, somapacitan has been shown to be efficacious and well tolerated with a prolonged half-life in plasma compared with recombinant growth hormone.^{24,25}

Using X-ray crystallography analysis, we identified only one specific binding site for somapacitan on HSA domain II at the low-affinity FA6, despite the identification of at least seven fatty acid-binding sites, distributed asymmetrically over the tertiary structure of HSA.^{1,26} The same binding site was seen for the ternary complex, where somapacitan was replaced by its albumin-binding side chain. This experiment was performed in parallel with the full-length somapacitan, because the flexibility of the albumin-binding side chain was anticipated to make crystallization experiments challenging. The flexibility of the side chain did hinder crystallization, but fortunately, there is a large enough cavity in the crystals to hold the GH component of somapacitan, enabling crystallization despite the absence of crystal contacts for GH. In our studies, a complex of an HSA analogue optimized for FcRn binding, HSAopt, and FcRn was first established before the HSAopt:FcRn:somapacitan ternary complex was generated. The crystal structure of the human FcRn:HSAopt complex was published by Schmidt et al. in 2013⁵ and recognized as a potential tool for the study of protracted binding of analogues to HSA by protein crystallography. The high solubility of HSA infers potential difficulties in crystallizing HSA in complex with molecules less soluble than HSA itself. Addition of FcRn to HSA both

reduced the solubility considerably and stabilized the otherwise rather flexible HSA by binding to both domains I and III.⁵

In the FcRn:HSAopt:somapacitan ternary complex, FcRn is bound to albumin on domains I and III. Fatty acid-binding sites on the two domains could therefore potentially be blocked by FcRn and prevent binding by somapacitan. This is highly relevant for this study as a high-affinity fatty acid-binding site on domain III, FAS, is located directly where the FcRn Trp59 residue interacts with HSA (reviewed by Sand et al.²⁶). Furthermore, a complex with a protracted derivative bound to the complex would provide insight into HSA binding compatible with simultaneous FcRn binding and hereby potentially allowing co-rescue of the protracted derivative together with HSA in the FcRn salvage pathway.²⁶ When generating the ternary complexes, we were guided by the following considerations. (i) It is important to consider the pH-dependent nature of the FcRn:HSA interaction. At pH >6.5, the complex is no longer formed. The binding strength is slightly improved in HSAopt, but the pH dependency still exists. (ii) Large amounts of sulfate should be avoided. Ammonium sulfate was used as the precipitating agent in the crystal structure determination of the HSAopt:FcRn complex by Schmidt et al.⁵ We have found that this concentration of sulfate blocks many of the fatty acid-binding pockets and prevents somapacitan binding (data not included). The crystal structure of the complex with unmodified HSA and FcRn has also been published,¹¹ and a comparison shows that the four mutated residues in HSAopt do not affect the overall complex structure, validating our choice of HSAopt for this study.

The formation of a complex of somapacitan and recombinant HSA was attempted by mixing the components in a 1:2 HSA:somapacitan molar ratio at room temperature, with the intention of removing excess somapacitan by SEC. Surprisingly, no excess somapacitan could be detected, suggesting an additional somapacitan-binding site on HSA. A mixture containing a 1:3 HSA:somapacitan molar ratio showed excess somapacitan, so at least two binding sites should be anticipated. The isolated HSA:somapacitan complex was also mixed with FcRn and subjected to SEC; in this case, FcRn could not remove somapacitan from HSA, emphasizing the importance of the mixing order when forming these complexes. ITC showed that somapacitan can bind HSA in the presence of FcRn, but the binding is much weaker when FcRn is present (data not shown). To identify the second site of binding of somapacitan on HSA, we studied the binding of somapacitan to two commercially available domain preparations of HSA. The stoichiometry of the HSA domain:somapacitan complexes

revealed one binding site on both the domain I+II and the domain III preparations, suggesting at least two binding sites for somapacitan on HSA. While the binding to domain I+II most likely corresponds to the site localized by the crystal structures, the binding to domain III would suggest a site previously blocked by the FcRn molecule in the ternary complex used for protein crystallography. On the basis of this knowledge, we propose a model of HSA with somapacitan bound at two sites: one at FA6 in domain II, as determined from the crystal structures, and one at FA4 or FA5 on domain III, deduced from SEC and ITC studies of both full-length HSA and its domains (Figure 6). This model also fits well with our SAXS data obtained on the HSA:somapacitan complex.

ITC analysis was used to further elucidate the properties of binding of somapacitan to HSA. Interestingly, the binding properties revealed a clear temperature dependency. A negative change in ΔC_p was observed for the high-affinity binding site and suggests binding dominated by hydrophobic interactions.¹³ Whereas a positive change in ΔC_p for the low-affinity binding sites indicates that forces other than just hydrophobic ones are involved in the binding. The results are in agreement with the current knowledge of the binding characteristics at FA4 and FA5 in domain III and FA6 in domain II: FA4 and FA5 are formed by hydrophobic channels in subdomains IIIA and IIIB, respectively, and FA6 is located at the interface between subdomains IIA and IIB, without a clear ligand for the carboxylate group of the fatty acids.^{1,6}

To confirm the location of the somapacitan-binding sites on HSA at FA4 or FA5 and FA6, titrations were performed using protein constructs of HSA domain III (containing FA4 and FA5) and domain I+II (containing FA6), respectively. The titration profile obtained for domain III was comparable to the profile of full length HSA, which supports the observations that the high-affinity binding site is located on domain III at FA4 or FA5. The titration profile of domain I+II indicated an interaction with somapacitan. Finally, we attempted to measure the kinetics of binding of HSA to somapacitan by SPR, but because there are multiple sites for binding of somapacitan on HSA, it was not possible to obtain reliable estimates for kinetic parameters. However, we were able to show that HSA binds somapacitan in a concentration-dependent manner and the binding kinetics for HSA and HSAopt are different (Figure 7).

CONCLUSIONS

In conclusion, we have used a combination of methods to identify a high-affinity binding site for somapacitan on HSA at FA4 or FA5, with a dissociation constant (K_D) of approximately 100–1000 nM using SEC and ITC techniques and a low-affinity binding site at FA6 with a K_D of approximately 1000–10000 nM using protein crystallography and ITC techniques. The low-affinity site (FA6) is compatible with FcRn-facilitated recirculation for extension of somapacitan circulation time, while the high-affinity FA4 or FA5 site cannot be used for this purpose, as it overlaps with the FcRn-binding site of HSA. This knowledge is valuable in understanding the pharmacology in using HSA for prolonged action of drug substances and may potentially be used for the engineering of HSA binding.

ASSOCIATED CONTENT

Accession Codes

FcRn, P55899; FcRn:HSAopt:albumin-binding side chain, 6QIP; FcRn:HSAopt:somapacitan, 6QIO; HSA residues 25–609, P02768; growth hormone, P61769.

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Author Contributions

E.J., A.D.N., H.D., and P.T. designed and planned the study. E.J. purified HSAopt, determined the crystal structures, and analyzed SEC and SAXS data. A.D.N. performed ITC analysis. H.D., C.B.S., and C.W. performed SPR analysis. T.H. and J.C. produced and purified FcRn. S.J. constructed the HSAopt-producing strain. J.P. produced the HSAopt. P.T. coordinated the work and wrote the manuscript together with E.J.

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Notes

The authors declare the following competing financial interest(s): All authors conducted the studies as employees of Novo Nordisk A/S and own stock in the company.

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ABBREVIATIONS

ΔC_p , change in heat capacity; FA, fatty acid-binding site; FcRn, neonatal Fc receptor; GH, growth hormone; GHD, growth hormone deficiency; HSA, human serum albumin; HSAopt, FcRn-binding optimized HSA; ITC, isothermal titration calorimetry; K_D , dissociation constant; MHC-class-I, major histocompatibility complex class-I; PDB, Protein Data Bank; SAXS, small-angle X-ray scattering; SEC, size exclusion chromatography; SPR, surface plasmon resonance; YPD, yeast extract peptone dextrose.

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